

ABSTRACT

This study evaluates National Water Model Version 3.0 (NWM 3.0) peak flow simulation accuracy at 101 USGS stream gauges across Oklahoma from 2000 to 2022 with 2,274 total event pairs. Gauges were classified as regulated (n=80) or unregulated (n=21) using a multi-criteria framework combining USGS peak flow qualification codes, GAGES-II dataset attributes, and NWIS drainage area ratios, and further divided by drainage area into small ($<1,000 \text{ km}^2$, n=66) and large ($\geq 1,000 \text{ km}^2$, n=35) basins. Various performance metrics like Kling-Gupta Efficiency (KGE), Nash-Sutcliffe Efficiency (NSE), Percent Bias (PBIAS), and Root Mean Square Error (RMSE) were computed for matched annual peak flow event pairs and compared using Mann-Whitney U, Welch's t-test, and permutation tests. NWM 3.0 showed marginal overall performance (median KGE = 0.29, NSE = -0.08 , PBIAS = $+10.08\%$). Unregulated gauges significantly outperformed regulated gauges (KGE = 0.51 vs. 0.24, $p < 0.01$), and small basins outperformed large basins (KGE = 0.38 vs. 0.28, $p < 0.001$). A sensitivity analysis also showed that these results were robust to selection of storm-window size.